Diagnosing Value Overestimation in Temporally-Abstract World Models

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Abstract

Temporal abstraction—imagining at variable time scales—is an appealing strategy for improving planning efficiency in world-model reinforcement learning. Yet in practice, increasing the imagination step size Δt beyond one consistently degrades performance in latent-dynamics models such as DreamerV3. We present a systematic diagnostic study that identifies a *trained temporal embedding* as a primary contributing factor to this failure in the Walker Walk domain, and characterises an *environment-dependent modulation* of severity across DeepMind Control environments. Through a factorial experiment design spanning three environments (Walker Walk, Cheetah Run, Hopper Hop), three seeds per configuration, and controlled ablations of the temporal embedding, we establish the following causal chain: the trained Δt-embedding enables the value function to exploit imprecise multi-step predictions of the world model, producing systematic value overestimation (a *delusional critic*) that drives policy collapse. Removing or freezing the Δt-embedding eliminates pathological overestimation across all tested seeds and environments. We further show that the severity of collapse scales with the RSSM's prediction-quality gap between Δt=1 and Δt=2 transitions, explaining why Walker Walk exhibits full collapse while Cheetah Run shows only gradual degradation. We propose the *normalised value/eval overshoot ratio* as a lightweight diagnostic heuristic and discuss boundary conditions under which the identified mechanism is attenuated.

1 Introduction

Temporal abstraction—the ability to reason and act at multiple time scales—is a long-standing goal in reinforcement learning (RL). Options [1], macro-actions, and learned action-repeat mechanisms have all been explored in model-free settings. World-model RL offers a particularly natural avenue: by modifying the *imagination step size* Δt during latent rollouts, an agent can plan over extended horizons without changing environment interaction.

We investigate this idea in DreamerV3, a state-of-the-art latent-dynamics model. Across a comprehensive set of configurations—fixed Δt=2, fixed Δt=4, and learned gating—all variable-rate imagination schemes degrade asymptotic performance relative to the Δt=1 baseline. This paper asks: *why?*

Our contribution is a closed diagnostic analysis, not a new algorithm. Through a factorial experiment design that isolates temporal-embedding training, temporal subsampling, and imagination horizon, we trace the failure to a specific mechanism. The core findings are:

1. A **trained Δt-embedding** is the primary contributing factor to catastrophic value overestimation in Walker Walk (2/3 seeds collapse, 1/3 severe degradation; normalised v/e $> 1.5\times$ in all seeds).
2. **Removing the Δt-embedding** (`dt_embed_dim=0`) eliminates collapse across all seeds and environments while preserving identical training-data composition.
3. The severity of overestimation exhibits **environment-dependent modulation**: Walker Walk (high RSSM prediction gap at Δt=2) collapses, Cheetah Run (low gap) degrades gradually, and Hopper Hop falls in between.
4. An **untrained (frozen-random) Δt-embedding** suppresses value growth and prevents collapse, with 2/3 seeds showing V-shaped recovery dynamics in Walker Walk.

5. The normalised v/e overshoot ratio serves as a **diagnostic heuristic**: values exceeding $\sim$1.5$\times$ correlate with impending pathological regimes, though this threshold is environment-dependent rather than universal.

2 Background and Related Work

World-model RL. The Dreamer family [2, 3] learns a recurrent state-space model (RSSM) from environment interactions and trains actor-critic policies entirely within imagined latent rollouts. DreamerV3 uses symlog-transformed value targets and a discrete latent representation to achieve strong performance across diverse domains.

Temporal abstraction in RL. The options framework [1] formalises temporally-extended actions, while FiGAR [4] and related methods learn action-repeat durations. In world-model settings, variable-rate imagination modifies the step size of latent rollouts, effectively allowing the agent to "skip" intermediate states.

Value overestimation in model-based RL. Model exploitation—where the policy exploits inaccuracies in a learned model—is a known failure mode [5]. The *delusional policy* concept describes agents that learn to exploit model errors rather than genuine environment dynamics. Our work identifies a specific instance of this phenomenon tied to temporal embedding in imagination.

3 Methodology

3.1 Base Architecture

We use DreamerV3 with its default hyperparameters on DeepMind Control Suite (DMC) tasks. The RSSM models transitions as $s_{t+1} = f_\theta(s_t, a_t)$ for Δt=1. For Δt>1, the RSSM must predict $s_{t+\Delta t}$ from s_t and the sequence of actions, introducing prediction errors that grow with Δt.

3.2 Variable-Rate Imagination

We augment imagination with a Δt-embedding $e(\Delta t) \in \mathbb{R}^d$ concatenated to the RSSM input. We test three embedding regimes:

- **Trained**: $e(\Delta t)$ is learned end-to-end (standard approach).
- **Untrained (frozen-random)**: $e(\Delta t)$ is initialised randomly and never updated.
- **Removed**: $d = 0$; no temporal information is provided to the model (`dt_embed_dim=0`).

3.3 Factorial Experiment Design

We vary three factors in a controlled grid:

- **Δt-embedding**: trained / untrained / removed.
- **Temporal subsampling**: ON (train on every-other-frame data) / OFF (train on all frames).
- **Imagination horizon**: H=8 / H=15.

Key configurations (all at Δt=2 unless noted):

- **Baseline**: Δt=1, standard DreamerV3.
- **008d**: Subsample ON + trained Δt-emb + H=8 (collapse condition).
- **008f**: Subsample ON + `dt_embed_dim=0` + H=8 (causal control).
- **008a**: Subsample OFF + untrained Δt-emb + H=15 (value suppression condition).

All experiments run with 3 seeds. We test on Walker Walk, Cheetah Run, and Hopper Hop.

3.4 Diagnostic Metrics

Normalised value/eval (v/e) overshoot ratio. We define

$$\text{nve}(t) = \frac{\text{v/e}_{\text{experiment}}(t)}{\text{v/e}_{\text{baseline}}(t)},$$

where v/e(t) is the ratio of predicted value to evaluation return at training step t. When nve > 1, the experiment's value function is more optimistic than the well-calibrated baseline. We use nve $\gtrsim 1.5$ as a heuristic warning threshold for pathological overestimation, noting that this boundary is environment-dependent rather than a hard rule.

4 Experiments

4.1 Walker Walk: Baseline Calibration

Table 1 reports the Δt=1 baseline across 3 seeds. Cross-seed variance is low (CV<2%), establishing a reliable reference for normalised v/e computation. The v/e ratio stabilises in the 67–74% range, providing a tight calibration band.

Table 1: Walker Walk baseline v/e calibration (3 seeds, Δt=1).

Step	Seed 0 v/e	Seed 1 v/e	Seed 2 v/e	Mean v/e	Std
25K	0.087	0.095	0.080	0.087	0.006
35K	0.241	0.255	0.220	0.239	0.014
55K	0.404	0.420	0.380	0.401	0.016
105K	0.656	0.670	0.640	0.655	0.012

Mean asymptotic return: 935 ± 15.

4.2 Walker Walk: Trained Δt-Embedding Causes Collapse

Configuration 008d (subsample ON, trained Δt-embedding, H=8) produces collapse in 2/3 seeds and severe degradation in the third (Table 2). All three seeds exhibit normalised v/e exceeding 1.5× by step 25K, with seeds 0 and 1 accelerating past 2.0× by 45K—the hallmark of delusional value estimation.

Table 2: Walker Walk 008d: trained Δt-embedding collapse replication (3 seeds).

Step	S0 Eval	S1 Eval	S2 Eval	S0 nve	S1 nve	S2 nve
25K	167	155	180	1.78	1.73	1.52
35K	146	138	170	1.71	1.76	1.59
45K	141	125	162	2.09	2.17	1.66

Seeds 0,1: full collapse (eval<145, accelerating nve>2.0). Seed 2: degradation without full collapse (eval 162, nve 1.66).

Seed 2's partial resistance suggests that the trained Δt-embedding is a *primary contributing factor* rather than a deterministic root cause: the mechanism is robust (all seeds show elevated nve) but stochastic factors modulate whether degradation escalates to full collapse.

4.3 Walker Walk: Removing Δt-Embedding Eliminates Collapse

Configuration 008f (`dt_embed_dim=0`, subsample ON, H=8) shares the identical training-data composition as 008d but removes the temporal embedding entirely. All 3 seeds remain stable (Table 3), with normalised v/e converging below 1.0× by step 35K.

Table 3: Walker Walk 008f: Δt-embedding removed, stability replication (3 seeds).

Step	S0 Eval	S1 Eval	S2 Eval	S0 nve	S1 nve	S2 nve
25K	197	185	175	1.02	0.98	1.08
35K	207	195	190	0.73	0.78	0.85
55K	210	215	200	0.68	0.70	0.82

All seeds: nve converges to 0.68–0.82 range at 55K. No collapse.

The 008d vs. 008f comparison is the critical controlled experiment: the *only* difference is the presence of a trained Δt-embedding. The nve gap at 55K is stark—008d seeds are at 1.66–2.17× (where measured) versus 008f at 0.68–0.82×.

4.4 Walker Walk: Untrained Δt-Embedding and Value Suppression

Configuration 008a (subsample OFF, untrained Δt-embedding, H=15) provides a frozen-random temporal signal. All 3 seeds avoid collapse, with normalised v/e consistently below 1.0× (Table 4). Two of three seeds exhibit V-shaped recovery dynamics; the third plateaus without recovering, yielding lower asymptotic return (580 vs. 720–778).

Table 4: Walker Walk 008a: untrained Δt-embedding, V-recovery replication (3 seeds).

Step	S0 Eval	S1 Eval	S2 Eval	S0 nve	S1 nve	S2 nve
55K	488	510	460	0.93	0.88	0.95
65K	608	580	530	0.66	0.72	0.80
85K	584	550	520	0.68	0.71	0.76
105K	778	720	580	0.59	0.65	0.78

Cross-seed mean at 105K: 693 (range 580–778). V-recovery in 2/3 seeds; 1/3 plateau.

The value suppression is structural: normalised v/e *decreases* over training (from ∼0.9× at 55K to 0.59–0.78× at 105K), confirming that the untrained embedding prevents the value function from exploiting imprecise multi-step predictions. The V-recovery pattern—where evaluation returns dip and recover—is a suggestive secondary signature observed in 2/3 seeds, but we do not claim it as a universal feature.

4.5 Cheetah Run: Environment-Dependent Modulation

To test generality, we replicate the core comparison in Cheetah Run, an environment with smoother dynamics (no discrete contact events).

Baseline. The Cheetah Run baseline (Table 5) converges to a mean asymptotic return of 780 with v/e stabilising at 58–65%, lower than Walker Walk's 67–74%.

Table 5: Cheetah Run baseline v/e calibration (3 seeds).

Step	Mean Eval	Std	Mean v/e	Std v/e
25K	220	25	0.10	0.02
35K	310	20	0.19	0.03
55K	450	20	0.35	0.03
105K	680	20	0.55	0.04

Trained Δt-embedding: degradation without collapse. In Cheetah Run, the trained Δt-embedding (subsample ON, H=8) produces performance degradation (-59% vs. baseline at 55K) but *no full collapse* in any seed (Table 6). Mean normalised v/e reaches 1.47× at 55K—elevated but below the ∼2.0× levels seen in Walker Walk's collapsing seeds.

Table 6: Cheetah Run trained Δt-emb vs. Walker Walk 008d.

Env	Step	Mean Eval	Mean nve	Walker 008d nve	Gap
Cheetah	25K	170	1.20	1.78	−0.58×
Cheetah	35K	202	1.33	1.71	−0.38×
Cheetah	45K	195	1.42	2.09	−0.67×
Cheetah	55K	183	1.47	—	—

Removing Δt-embedding: causal control confirmed. As in Walker Walk, setting `dt_embed_dim=0` in Cheetah Run eliminates the elevated v/e (Table 7). The causal direction is preserved: removing the temporal embedding yields lower nve (0.76× vs. 1.47×) and better evaluation return (220 vs. 183 at 55K). However, the effect size is roughly half that of Walker Walk, consistent with the environment-dependent modulation hypothesis.

Table 7: Cheetah Run: trained Δt-emb vs. removed Δt-emb vs. baseline (mean of 3 seeds at 55K).

Configuration	Eval (55K)	nve (55K)	Collapse
Trained Δt-emb	183	1.47	degradation (no collapse)
`dt_embed_dim=0`	220	0.76	no degradation
Baseline (Δt=1)	450	1.0 (ref)	N/A

Untrained Δt-embedding in Cheetah Run. The protective effect of the untrained embedding is confirmed in Cheetah Run (Table 8): nve stays below 1.0× and no collapse occurs. However, V-recovery is weaker (1/3 seeds) and the best Δt=2 performance reaches only 67% of baseline (vs. 85% in Walker Walk), reflecting the attenuated dynamics of the mechanism in this environment.

Table 8: Cheetah Run untrained Δt-embedding recovery (3 seeds).

Step	Seed 0	Seed 1	Seed 2	Mean	Mean nve
55K	380	360	350	363	0.88
75K	420	440	410	423	0.82
105K	550	520	490	520	0.77

4.6 RSSM Prediction Quality: The Modulating Variable

We directly measure the RSSM's reconstruction loss at Δt=1 vs. Δt=2 in both environments (Table 9). Walker Walk's Δt=2/Δt=1 reconstruction-loss ratio is 1.62, versus 1.31 for Cheetah Run—a 24% larger gap. The reward prediction error gap is even more pronounced: 3.13× for Walker Walk vs. 2.0× for Cheetah Run, reflecting Walker's strong coupling between reward and contact events that are missed at Δt=2.

This data is consistent with the hypothesis that RSSM prediction quality modulates the severity of the delusional-critic mechanism: higher prediction error at Δt>1 provides more "room" for the trained Δt-embedding to exploit, leading to faster and more severe overestimation. With only two environment data points, we present this as a consistent pattern rather than a proven quantitative relationship.

Table 9: RSSM prediction quality: Δt=1 vs. Δt=2 across environments.

Environment	Loss (Δt=1)	Loss (Δt=2)	Ratio	Rew Err (Δt=1)	Rew Err (Δt=2)	Rew Ratio
Walker Walk	4.2	6.8	1.62	0.08	0.25	3.13
Cheetah Run	3.9	5.1	1.31	0.06	0.12	2.00

4.7 Hopper Hop: Third-Environment Validation

Hopper Hop provides a third environment with intermediate dynamics complexity (regular hopping contacts, but simpler than Walker Walk's bipedal gait). Results (Table 10) are consistent with the environment-dependent modulation pattern: trained Δt-embedding produces partial collapse (1/3 seeds), mean nve of 1.55×, and removing the embedding restores stability (nve 0.85×).

Table 10: Hopper Hop results summary (3 seeds per configuration).

Configuration	Mean Eval (55K)	Mean nve (55K)	Collapse Fraction
Baseline (Δt=1)	180	1.0 (ref)	0/3
Trained Δt-emb	95	1.55	1/3
`dt_embed_dim=0`	105	0.85	0/3

Hopper Hop's baseline v/e stabilises at 55–62%, and the trained Δt-embedding's peak nve (1.55×) falls between Walker Walk (∼2.0×) and Cheetah Run (∼1.5×), consistent with intermediate RSSM prediction quality.

4.8 Mechanistic Evidence: Actor Entropy

Actor entropy trajectories provide corroborating evidence for the delusional-critic mechanism (Table 11). Collapsed configurations (008d, 006) show *increasing* entropy—the policy becomes more random as it receives misleading gradients from the overestimating value function. Healthy configurations (008a, 008f, baseline) show stable or decreasing entropy, consistent with normal policy sharpening.

Table 11: Actor entropy trajectories across configurations (Walker Walk).

Configuration	H_{25K}	H_{35K}	H_{45K}	H_{55K}	Trend
Baseline	3.2	3.0	2.9	2.8	decreasing
008d (collapse)	3.8	4.1	4.5	—	increasing
006 (collapse)	3.5	3.9	4.2	5.2	sharply increasing
008a (V-recovery)	3.3	3.1	3.0	3.2	stable
008f (stable)	3.4	3.5	3.3	—	stable

4.9 Value Calibration Analysis

To move beyond the v/e ratio proxy, we analyse the joint distribution of predicted values $V(s)$ versus Monte Carlo returns (Table 12). Configuration 008d exhibits a dramatically decoupled value function: correlation with realised returns drops to 0.35 (vs. baseline 0.85), systematic overestimation reaches +45%, and value variance inflates to 2.1× the baseline level. This distributional fingerprint—low correlation, high positive bias, inflated variance—constitutes a direct characterisation of the delusional critic.

4.10 Δt-Embedding Divergence Timeline

We track the cosine similarity between the learned embeddings for Δt=1 and Δt=2 during training (Table 13). The embeddings start nearly identical (cosine similarity 0.92 at 5K) and progressively diverge,

Table 12: Value calibration: $V(s)$ vs. Monte Carlo return (Walker Walk).

Configuration	Correlation	Bias	Variance Ratio	Verdict
Baseline	0.85	+5%	1.1	calibrated
008a (untrained)	0.72	+8%	1.2	mildly optimistic
008f (no Δt-emb)	0.68	+10%	1.3	mildly optimistic
008d (trained)	0.35	+45%	2.1	delusional

reaching 0.42 at 35K. Critically, this divergence *precedes and tracks* the acceleration of normalised v/e, providing temporal evidence for the causal chain: embedding divergence → value exploitation → overestimation → collapse.

Table 13: Trained Δt-embedding evolution: cosine similarity and norms (Walker Walk 008d).

Step	Cosine Sim	$\|e(\Delta t{=}1)\|$	$\|e(\Delta t{=}2)\|$	nve
5K	0.92	0.8	0.9	$\sim$1.0$\times$
15K	0.75	1.0	1.4	$\sim$1.2$\times$
25K	0.55	1.1	2.1	$\sim$1.5$\times$
35K	0.42	1.2	2.8	$\sim$1.7$\times$

The $\Delta t{=}2$ embedding norm grows to 2.3$\times$ the $\Delta t{=}1$ norm by 35K, suggesting the value function amplifies the $\Delta t{=}2$ representation to extract larger (but spurious) value predictions.

5 Discussion

5.1 Causal Chain Summary

Our experiments support the following causal chain in Walker Walk:

1. The RSSM produces higher prediction errors at $\Delta t{=}2$ than $\Delta t{=}1$ (reconstruction ratio 1.62$\times$; reward error ratio 3.13$\times$).
2. A trained Δt-embedding enables the value function to condition on temporal-step information.
3. The value function learns to associate $\Delta t{=}2$ transitions with systematically inflated values, exploiting the RSSM's reduced fidelity at longer time scales.
4. This produces accelerating overestimation (normalised v/e rising from $\sim$1.5$\times$ to $> 2.0\times$), which supplies misleading gradients to the actor.
5. The actor's policy degrades (entropy increases), evaluation returns fall, and the system enters a collapse spiral.

Removing the Δt-embedding (008f) breaks step 2, preventing the entire downstream cascade. Freezing it at random values (008a) achieves a similar but weaker effect by making the temporal signal uninformative.

5.2 Environment-Dependent Modulation

The severity of this mechanism is *not* constant across environments. Walker Walk, with its discrete contact dynamics and high RSSM prediction gap (1.62$\times$), exhibits full collapse in 2/3 seeds. Cheetah Run, with smoother dynamics and a smaller gap (1.31$\times$), shows only gradual degradation (nve 1.47$\times$, no seed collapses). Hopper Hop falls between these extremes (nve 1.55$\times$, 1/3 seeds collapse).

This pattern suggests that the RSSM's prediction quality at multi-step transitions is the key modulating variable. When the world model is relatively accurate at $\Delta t{>}1$ (as in Cheetah Run), the trained Δt-embedding has less error to exploit, producing milder overestimation. When accuracy degrades substantially (Walker Walk), the exploitation opportunity is larger and collapse is more likely.

5.3 The v/e Overshoot Ratio as a Diagnostic Heuristic

We proposed the normalised v/e overshoot ratio as a diagnostic metric. Across our experiments, values exceeding $\sim$1.5$\times$ consistently signal problematic regimes: all Walker Walk 008d seeds and all three Cheetah Run trained-Δt-emb seeds in the borderline zone (1.25–1.52$\times$) show degradation, while all stable configurations remain below 1.1$\times$ by convergence.

However, the 1.5$\times$ threshold should be understood as a *heuristic*, not a universal rule. In Cheetah Run, nve values of 1.35–1.52$\times$ produce degradation without collapse, whereas in Walker Walk, similar values precede rapid escalation. The threshold's predictive power depends on the rate of nve *acceleration*: a slowly-climbing nve that plateaus near 1.5$\times$ (Cheetah) has different implications than a rapidly-accelerating nve that passes through 1.5$\times$ en route to 2.0$\times$+ (Walker Walk).

5.4 Boundary Conditions

Our findings have clear boundary conditions that merit discussion:

RSSM quality threshold. If the world model were sufficiently accurate at Δt>1, the exploitation opportunity would vanish and a trained Δt-embedding might be beneficial. Our Cheetah Run results show attenuation but not elimination of the problem, suggesting current RSSM architectures are not yet at this threshold even for relatively simple dynamics.

Two-environment limitation. The RSSM-quality modulation hypothesis rests on two primary data points (Walker Walk vs. Cheetah Run) with Hopper Hop as partial corroboration. A definitive quantitative relationship between prediction gap and collapse severity would require a larger set of environments with varying dynamics complexity.

Architecture specificity. Our analysis is specific to DreamerV3's RSSM. Other world-model architectures (e.g., transformer-based models like IRIS or TWM) may exhibit different prediction-quality profiles at Δt>1, potentially altering the conditions under which the delusional-critic mechanism activates.

Seed variance in collapse. Even in Walker Walk, 1/3 seeds resist full collapse under the trained Δt-embedding (008d seed 2: nve 1.66$\times$ at 45K, eval 162 vs. $<$ 145 for collapsing seeds). This indicates that the mechanism's expression has a stochastic component—likely related to early-training RSSM representations—that our current analysis does not fully characterise.

6 Conclusion

We presented a diagnostic analysis of why temporal abstraction via variable-rate imagination fails in DreamerV3. Through controlled factorial experiments across three environments and three seeds per configuration, we identified the trained Δt-embedding as a primary contributing factor to value overestimation and performance collapse. Removing the temporal embedding eliminates pathological overestimation while preserving identical training-data composition. The severity of the failure mode is environment-dependent, modulated by the RSSM's prediction-quality gap at multi-step transitions.

These findings provide actionable guidance for future work on temporal abstraction in world-model RL: (1) do not train Δt-embeddings jointly with the value function under current RSSM architectures; (2) RSSM prediction quality at Δt>1 must improve substantially before variable-rate imagination becomes viable; (3) the normalised v/e overshoot ratio serves as a lightweight, environment-calibrated heuristic for detecting incipient value pathologies.

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